

PROJECT DESIGN PHASE-II: SOLUTION REQUIREMENTS

Project Name: Auto Ticket Classification using Flow Design	Maximum Marks: 4 Marks
Date: 01 october 2026	Team ID: SWTID-2026-1536

FUNCTIONAL REQUIREMENTS (EPIC & STORIES)

FR NO.	FUNCTIONAL REQUIREMENT (EPIC)	SUB REQUIREMENT (STORY / SUB-TASK)
FR-1	Ticket Ingestion & Parsing	Ingestion of support tickets via Email Gateway / API Parsing unstructured text and extracting metadata (Subject, Body, Priority)
FR-2	Automated Classification Flow	NLP-based intent detection and sentiment analysis Rule-based and ML-driven category tagging (Technical, Billing, General)
FR-3	Smart Routing & Assignment	Dynamic assignment based on agent workload and skill matrix Escalation workflow trigger for high-priority or SLA-breach-tickets
FR-4	Feedback Loop & Analytics Dashboard	Agent feedback interface for misclassified ticket correction Real-time dashboard reporting ticket volume, accuracy metrics, and routing latency

NON-FUNCTIONAL REQUIREMENTS

FR NO.	NON-FUNCTIONAL REQUIREMENT	DESCRIPTION
NFR-1	Usability	Intuitive user interface with clear visual indicators for ticket queues, and accessible navigation for support agents and administrators.
NFR-2	Security	End-to-end encryption for data in transit (TLS 1.3) and at rest (AES-256). Role-Based Access Control (RBAC) adhering to data privacy standards.
NFR-3	Reliability	System availability of 99.9% uptime with automated failover mechanisms and robust error logging to prevent ticket processing loss during network disruptions.

FR NO.	NON-FUNCTIONAL REQUIREMENT	DESCRIPTION
NFR-4	Performance	Ticket classification and routing decision execution within under 2 seconds per request, supporting high-throughput parallel ingestion queues during peak traffic hours.
NFR-5	Availability	24/7 continuous operation with automated health checks, self-healing container pods, and scheduled zero-downtime maintenance windows.
NFR-6	Scalability	Horizontal auto-scaling capabilities designed to seamlessly handle seasonal spikes in support ticket volume without performance degradation or latency increase.